
Exercise sheet 3 for Numerical Optimization

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Exercise 1 (Second-order necessary optimality condition, 7+2+2 points)

- a) Prove theorem 3.1.6 in the lecture notes. That is if $f \in C^2(\Omega, \mathbb{R})$ on an open set $\Omega \subset \mathbb{R}^n$ attains a local minimum at $\hat{x} \in \Omega$, then it holds that $\nabla f(\hat{x}) = \mathbf{0}$ and $H_f(\hat{x})$ is positive semidefinite.

Note: A matrix $A \in \mathbb{R}^{n \times n}$ is positive semidefinite if

$$v^T A v \geq 0 \quad \forall v \in \mathbb{R}^n .$$

It is positive definite if

$$v^T A v > 0 \quad \forall v \in \mathbb{R}^n \setminus \{\mathbf{0}\} .$$

If $f \in C^2(\Omega, \mathbb{R})$ satisfies $\nabla f(\hat{x}) = \mathbf{0}$ at $\hat{x} \in \Omega$ the following two statements hold:

- Theorem 3.1.5: If $H_f(\hat{x})$ is positive definite, then f has a strict local minimum at $\hat{x}$.
- Theorem 3.1.6: If f has a local minimum at $\hat{x}$, $H_f(\hat{x})$ is positive semidefinite.

We want to find examples in order to see that the statements' converses are not true.

- b) Find a C^2 -function f and a point $\hat{x}$ with $\nabla f(\hat{x}) = \mathbf{0}$ that has a strict local minimum at $\hat{x}$, but $H_f(\hat{x})$ is not positive definite.
- c) Find a C^2 -function f and a point $\hat{x}$ with $\nabla f(\hat{x}) = \mathbf{0}$ and positive semidefinite $H_f(\hat{x})$, but f does not have a local minimum at $\hat{x}$.

Exercise 2 (Convex sets, 2+6 points)

A set $K \subset \mathbb{R}^n$ is called *convex* if

$$x, y \in K, \alpha \in [0, 1] \implies \alpha x + (1 - \alpha)y \in K . \quad (1)$$

a) Draw a two-dimensional convex set and a two-dimensional non-convex set and explain.

b) Let $x_1, \dots, x_m \in \mathbb{R}^n$, then

$$x = \sum_{i=1}^m \lambda_i x_i$$

is a *convex combination* of $x_1, \dots, x_m$, if $\lambda_i \geq 0$ for $i = 1, \dots, m$ and $\sum_{i=1}^m \lambda_i = 1$.

Prove that a set is convex if and only if it contains all convex combinations of its elements.

Note: Prove *convex* $\Rightarrow$ *contains all convex combinations* using induction over m .

Exercise 3 (Convex functions, 2+2+3+4+4 points)

Let $K \subset \mathbb{R}^n$ be convex and $K \neq \emptyset$. A function $f : K \rightarrow \mathbb{R}$ is *convex* on K , if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in K$ and $\lambda \in [0, 1]$. It is *strict convex* if

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in K$ with $x \neq y$ and $\lambda \in (0, 1)$.

- a) Why does K have to be convex in order to define a convex function $f : K \rightarrow \mathbb{R}$?
- b) Find a function mapping from $\mathbb{R}$ to $\mathbb{R}$ that is convex and draw it. Explain, how convexity can be illustrated in this simple example.
- c) Let $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Show that the affine function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = a^T x + b$$

is convex.

- d) Let $Q \in \mathbb{R}^{n \times n}$ be symmetric and $p \in \mathbb{R}^n$, $q \in \mathbb{R}$. Show that the quadratic function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \frac{1}{2}x^T Qx + p^T x + q$$

is convex if Q is positive semidefinite.

- e) Let $a \in \mathbb{R}^n$. Prove that

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \|x - a\|_2$$

is convex, but not strictly convex.

Exercise 4 (Compositions of convex functions, 4+4+4+4 points)

Let $C \in \mathbb{R}^n$ be convex. Prove the following statements about compositions of convex functions.

a) Let $f_i : C \rightarrow \mathbb{R}$ be convex and $\beta_i \geq 0$ for $i = 1, \dots, m$.

i) Show that

$$g(x) := \sum_{i=1}^m \beta_i f_i(x)$$

is convex on C .

ii) Additionally, suppose that for an arbitrary but fixed $k \in \{1, \dots, m\}$, f_k is strictly convex and $\beta_k > 0$. Prove that then $g(x)$ is strictly convex on C .

b) For an affine function

$$g : C \rightarrow \mathbb{R}^m, \quad g(x) := Ax + b \quad \text{with} \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m$$

show that

i) the set $g(C)$ is convex.

ii) Also show that if $f : g(C) \rightarrow \mathbb{R}$ is convex, $f \circ g$ is also convex on C .

c) For $f : C \rightarrow \mathbb{R}$ convex, show that

$$g(x) := \max\{0, f(x)\}$$

is also convex on C .

d) For $f : C \rightarrow \mathbb{R}$ convex and non-negative, show that

$$g(x) := f(x)^2$$

is also convex on C .